runDIC quick start

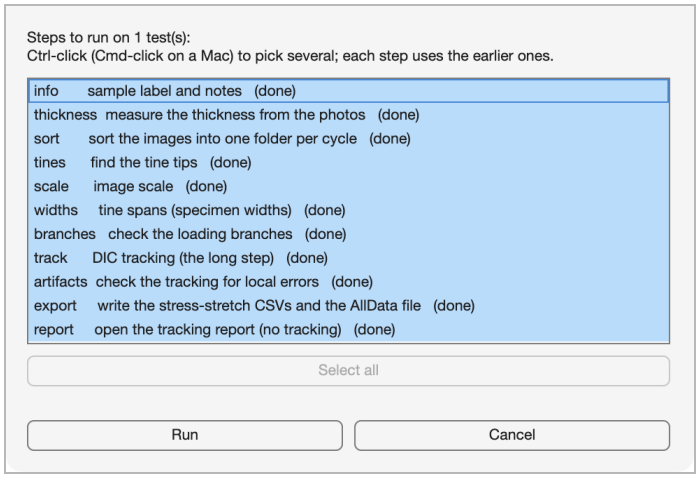
CellScale biaxial test → stress–stretch data, in MATLAB. The full guide is `README.md` in the `matlab` folder.

Before the first run

- MATLAB R2021a or later; no toolboxes needed.
- Copy the `matlab` folder to your computer (for example `Documents\runDIC`) and make it MATLAB's current folder (the folder bar at the top).
- The data: a sample folder (e.g. `N1900_LFPA`) holding one folder per test, named `test1_…`, `test2_…`, with the tester's files as it wrote them, and the photos of the strips in `Thickness, <ID>\test <N>` (e.g. `Thickness, 1900\test 1`).

Start

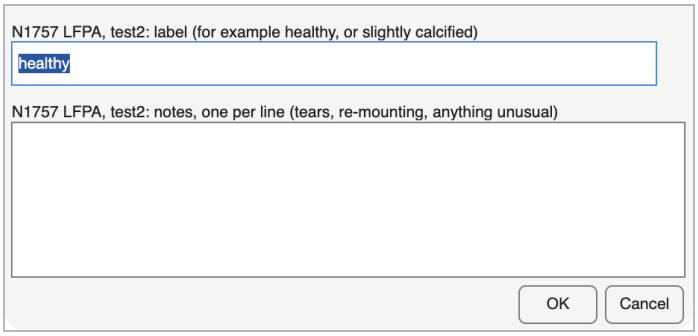
Type `runDIC` in the Command Window and choose the **test folder** (or the sample folder, to run all its tests). The step list opens: keep the checked steps and click **Run**. Then answer each window as below; a double-click on the last point of a step does the same as its first button.



The step list. "(done)" marks what earlier runs did.

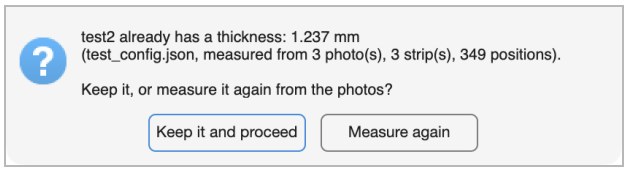
1 Label and notes

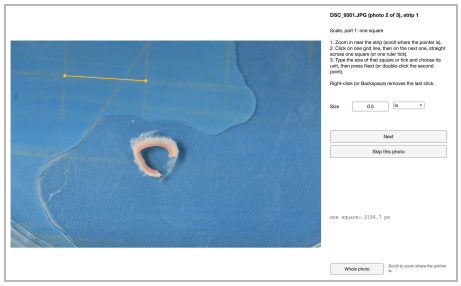
Type the sample's label (for example *healthy*) and note anything unusual: a tear, re-mounting, a bubble. Click OK.



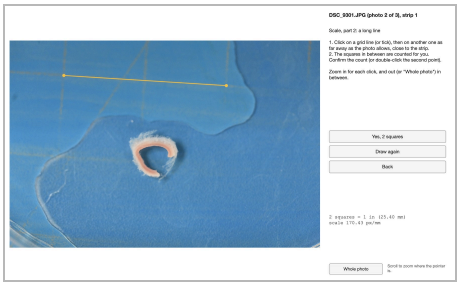
❷ Thickness from the photos

Each photo opens in turn. Scroll to zoom where the pointer is; right-click (or Backspace) removes the last click. If the test already has a thickness, you are asked first; choose **Keep it and proceed** unless you are measuring it again.

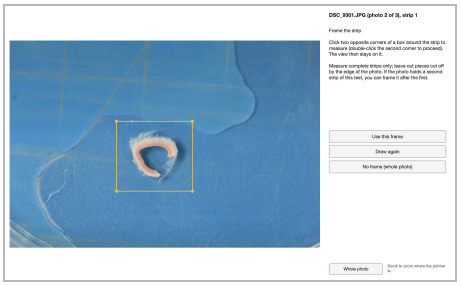




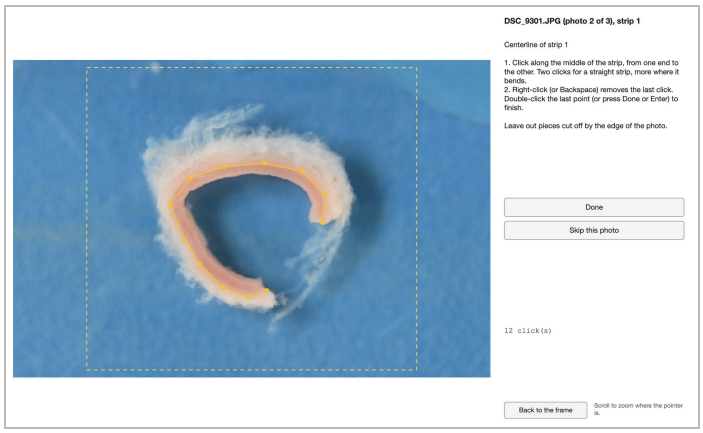
2a Scale, one square. Click on one grid line, then on the next one, straight across one square. Type its size (a mat square is **0.5 in**), press Next.



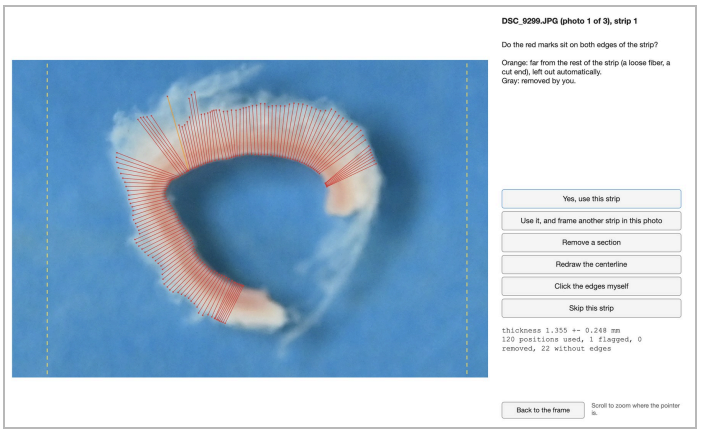
2b Scale, long line. Click on two grid lines as far apart as the photo allows, near the strip. Check the square count and confirm it. If the ends miss the grid lines, the window says so: **Draw again**.



2c Frame. Click two opposite corners of a box around one complete strip, then **Use this frame**. Skip pieces cut off by the photo's edge.



2d Centerline. Click along the middle of the strip, end to end (more clicks where it bends), then **Done**.



2e Check the edges.

> In 2e, the red lines should run across the strip and end on both of its edges. If they do, click **Yes, use this strip** (the next photo opens). Orange lines are left out automatically.

If not: **Remove a section** (click its two ends) for a bad stretch, **Redraw the centerline**, or **Click the edges myself** (click along one edge, then the other). **Skip this strip** if nothing works. A strip needs at least 10 positions; on a poor photo, redraw the centerline or click the edges yourself.

If no strip could be measured (or you close the window), a box asks you to type the thickness in mm; leave it blank and the stresses are left empty. A test without thickness photos skips this step, and a window at the start of the run asks for the thickness instead.

Checks before tracking

3 Sorting check

Every cycle (Def01–Def19) with its images and the frames missing from the tester's files. With no red rows, click **Yes, proceed**. A few missing frames (a red row with 3–5 missing) are fine: the camera drops frames now and then, so also **Yes, proceed**. A folder with no images, or more missing, means something went wrong: click **No, stop this test** and ask your supervisor.

Folder	Images	First image (s)	Last image (s)	Images missing	Data.csv rows missing	Coords1.csv rows missing	Note
Def01_Preconditioning_c01	300	0.0	59.8	0	0	0	
Def02_Preconditioning_c02	300	60.0	119.8	0	0	0	
Def03_Preconditioning_c03	300	120.0	179.8	0	0	0	
Def04_Preconditioning_c04	300	180.0	239.8	0	0	0	
Def05_Preconditioning_c05	300	240.0	299.8	0	0	0	
Def06_Preconditioning_c06	300	300.0	359.8	0	0	0	
Def07_Preconditioning_c07	300	360.0	419.8	0	0	0	
Def08_Preconditioning_c08	300	420.0	479.8	0	0	0	
Def09_Preconditioning_c09	300	480.0	539.8	0	0	0	
Def10_Preconditioning_c10	300	540.0	599.8	0	0	0	
Def11_Equibiaxial	300	600.0	659.8	0	0	0	
Def12_1-0.75_ratio	226	660.0	704.9	0	0	0	
Def13_1-0.5_ratio	150	705.0	734.8	0	0	0	
Def14_1-0.25_ratio	140	735.0	762.8	0	0	0	
Def15_Equibiaxial	300	763.0	822.8	0	0	0	
Def16_0.75-1_ratio	300	823.0	882.8	0	0	0	
Def17_0.5-1_ratio	300	883.0	942.8	0	0	0	
Def18_0.25-1_ratio	300	943.0	1002.8	0	0	0	
Def19_Equibiaxial	301	1003.0	1063.0	0	0	0	

test2 10062026 PLC25 30-14: 5317 images, one every 0.20 s; 0 images outside every cycle (before the first or after the last).

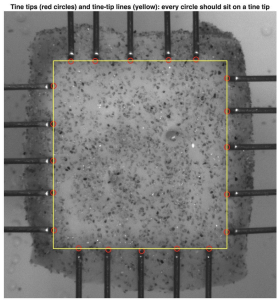


Every folder has all its images, Data.csv rows and Coords1.csv rows.

Proceed with this test?

Yes, proceed | No, stop this test

4 Tine tips



Every red circle should sit on the tip of a tine. If so, **Yes, continue**.

If not, **No, fix some rakes**, pick the wrong rakes and click the 5 tips of each (the end of each rod, where it meets the tissue); right-click removes the last click. The question then comes back with the new tips.

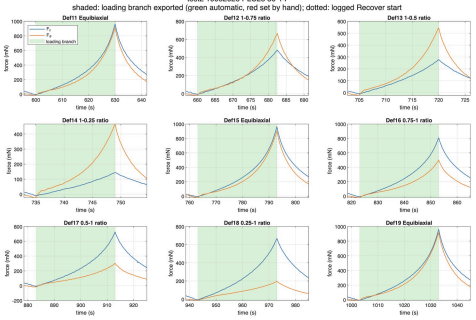
If the tips of a rake can't be found at all, a window asks you to click its 5 tips right away.



Is every red circle on the tip of a tine?
(The yellow rectangle joins the tips of each rake.)

Yes, continue | No, fix some rakes

5 Loading branches



Each green band should start where the forces begin to rise and end at their peak. If so, **Yes, continue**.

If one doesn't, **No, fix some**, pick it and click where loading starts and where it ends. Right-click removes the last click; the **A** key goes back to the automatic band.



Does each shaded band cover one loading curve,
from where the forces start to rise to where they peak?

Yes, continue | No, fix some

Tracking and the report

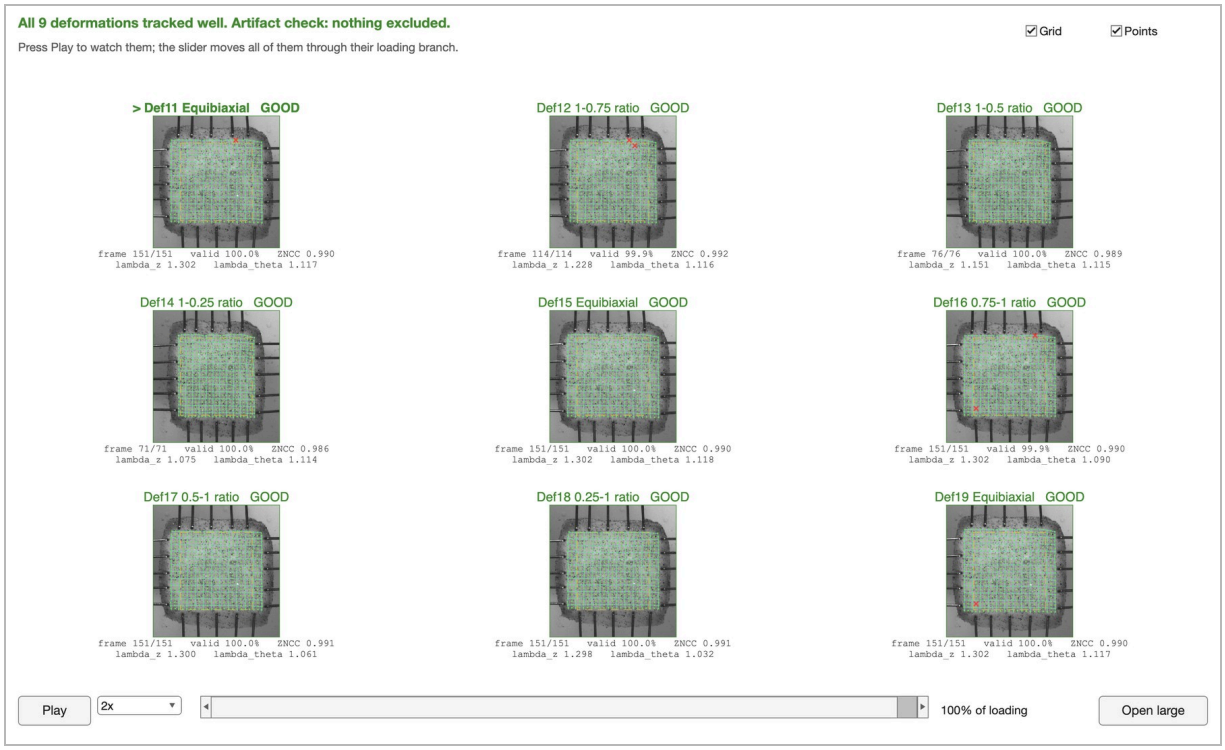
6 Tracking

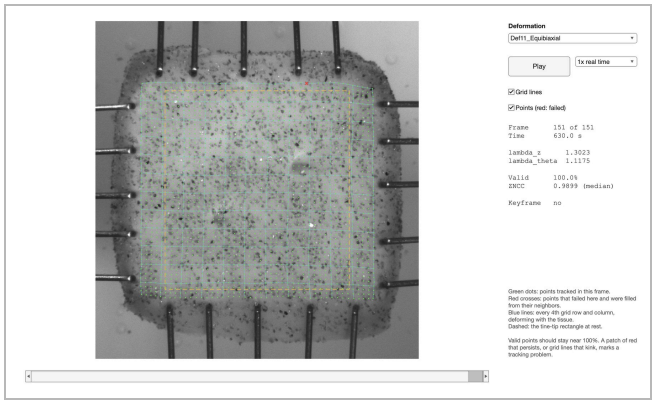
The run now proceeds by itself, about 15 minutes per test (longer on a laptop). Each deformation's progress line updates in the Command Window:

```
[1/19] Def01_Preconditioning_c01: 151 frames (loading)
  Def01  151/151 frames  100%  0m47s elapsed, 0m00s left  (valid 99.5%, ZNCC 0.9895)
  Def01_Preconditioning_c01 done: valid per frame median 99.7%, min 99.4%; bridge valid 100.0%
[2/19] Def02_Preconditioning_c02: 151 frames (loading)
  Def02  151/151 frames  100%  0m44s elapsed, 0m00s left  (valid 99.8%, ZNCC 0.9893)
  Def02_Preconditioning_c02 done: valid per frame median 99.9%, min 99.7%; bridge valid 100.0%
[3/19] Def03_Preconditioning_c03: 151 frames (loading)
  Def03  116/151 frames   77%  0m32s elapsed, 0m10s left  (valid 99.7%, ZNCC 0.9912)
```

7 Tracking report

At the end, the report shows Def11–Def19 side by side; **Play** runs them. Green dots are tracked points, red crosses failed ones. Each tile is marked **GOOD**, **CHECK** or **PROBLEM**; **Open large** shows the selected one big.





"Open large": one deformation, frame by frame.

> **Stop and ask your supervisor** before using the results when:
>
> - the sorting table has a folder with no images, or more than 5 frames missing;
> - tine tips or a loading branch can't be put right by clicking;
> - the thickness edges stay off the strip;
> - the Command Window says an artifact was excluded, or the report has any CHECK or PROBLEM tile.

If something goes wrong

"… excluded … tracking again" in the Command Window	a local tracking error was found and left out; the test is tracked a second time, so the run takes about twice as long. Expected; tell your supervisor.
`=== test… FAILED: <message>`	that test stopped; write down the message and ask your supervisor. With a sample folder, the other tests still run.
Stopping a run	press **Ctrl-C** in the Command Window. Run `runDIC` again: the step list marks the steps already done; select the rest. A tracking step that was interrupted starts over.

The results

In the test's `stress_stretch` folder:

`stress-stretch test<N>.AllData.xlsx`	for your modeling script: the layout of CellScale's AllData export, with the DIC stretches and stresses
`stress_stretch.png`	every protocol on one plot, to check at a glance
`all_protocols.csv`, `DefNN_….csv`	the same data as CSV files
`README.txt`	what was used: thickness, widths, scale, tracking verdict

Common tasks

Open the tracking report again	type `viewTracking` and choose the test folder
Change the thickness, without tracking again	`runDIC`, choose the folder, select only **thickness** and **export**
Redo a step	`runDIC`, select that step and the ones after it